# AMD-STEM Presentation 2025-2026

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# Why are we here?

Massachusetts STEM Week Oct 20-24, 2025
National Engineers Week Feb 22-28, 2026
Global Engineering Day Mar 4, 2026
National Science and Engineering Week Mar 6-15, 2026
National Engineering Month March 2026

We want to:

- Tell you about AMD
- Talk about engineering and what engineers do
- Get you excited about a career in science & engineering
- Give you a challenge to solve

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# What does Advanced Micro Devices do?

Develops Processors for Computers, Gaming Consoles, Smartphones, AI products, and even the Formula 1 team!

Microsoft Xbox360 and all Xbox One since

Samsung Exynos, Galaxy S24 phone

Sony PlayStation

Sponsors Mercedes Formula1 Team



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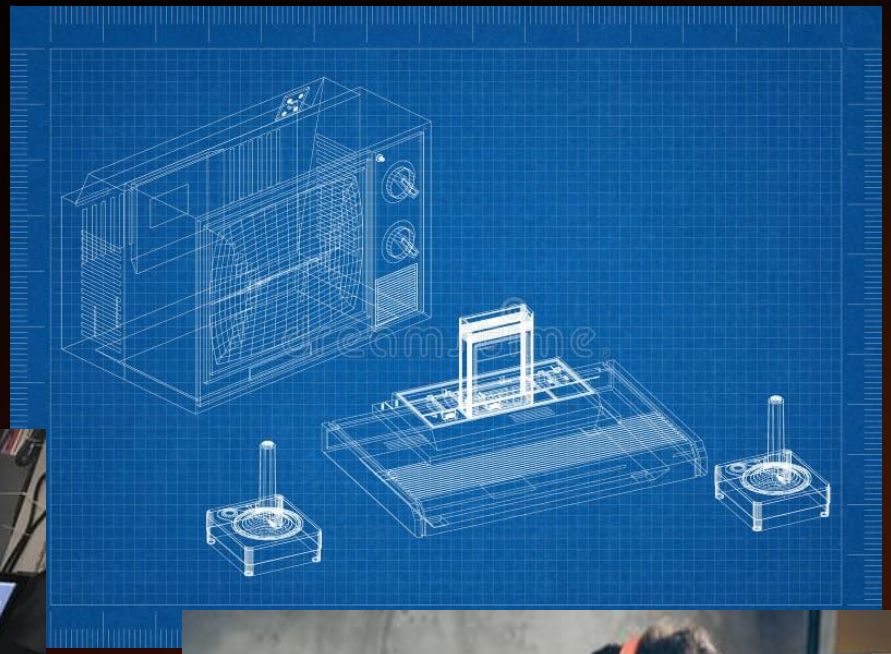
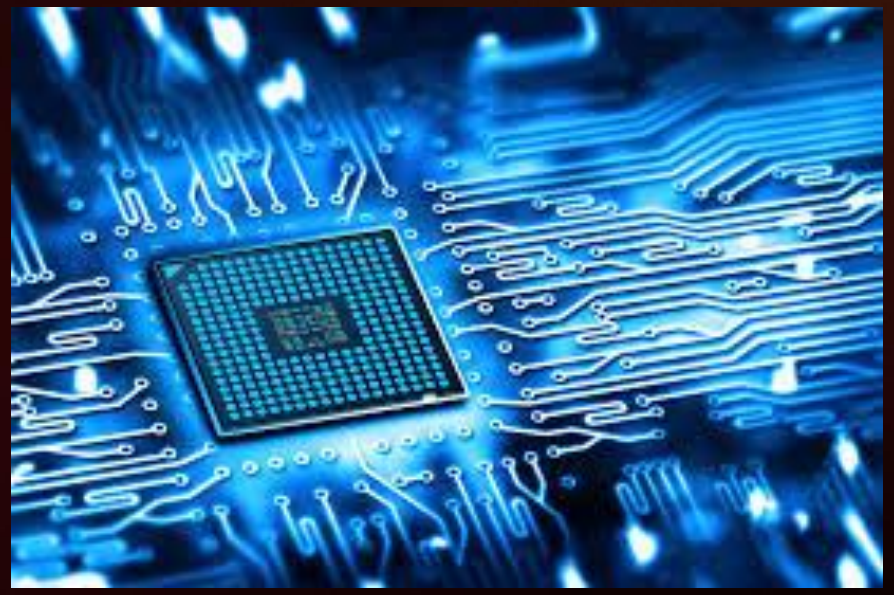
# SO -

…What do you think Engineers do?



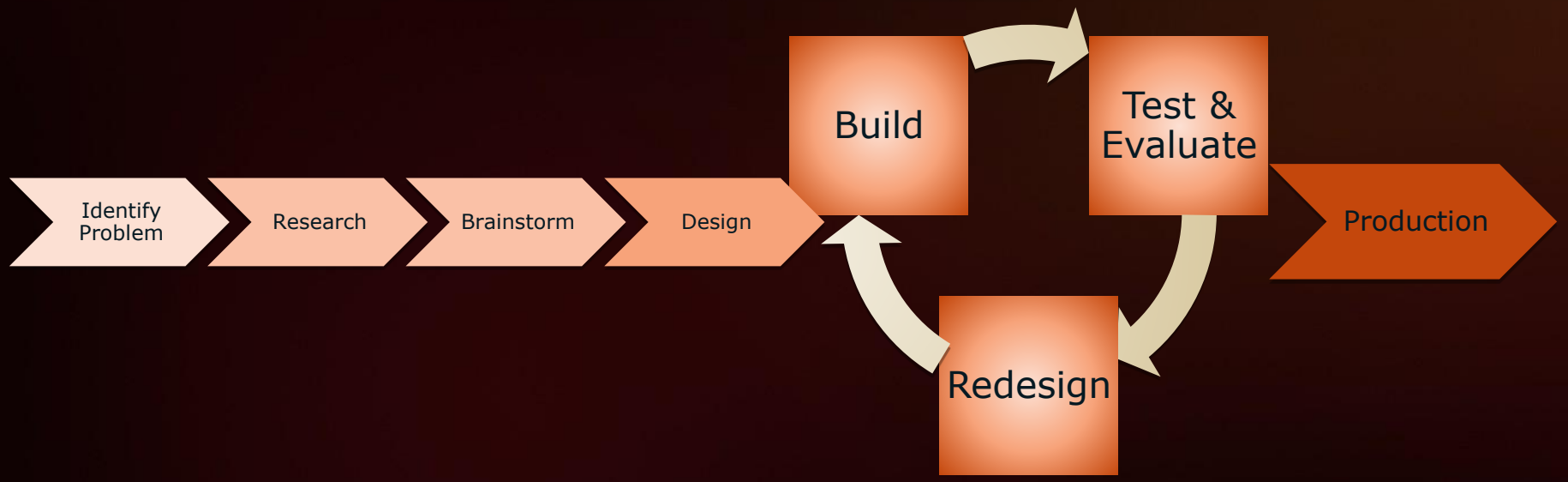
# Engineers…

…work in teams to **solve problems** to make the world a **better place**.



# How do Engineers work?

They follow the Design Process



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# Really, What do Engineers do?

- Find solutions to difficult problems
- Work in groups
- Make life better for everyone

## What tools and skills do we use

- We follow the design method
- We draw diagrams and we write descriptions of how we want to solve the problem
- We write code to implement our solutions, build circuits and machines
- We test our solutions
- If our solutions fail, we go back to debugging and redesigning our solutions
- When we succeed, we publish our solutions → products
- Get paid for solving problems!



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# Questions?

# Now it’s Time for Your Challenge

- Problem Statement:
  - Design a system that detects the distance of an object and indicates how close the object is by lighting up different colors

- Tools at your disposal:
  - Arduino controller board
  - Code development tool
  - Various electronic components
  - Reference guides for developing your solution
- Skills to use:
  - Work in teams
  - Discuss problem statement and propose solutions to your teammates
  - Sketch out your final solution on paper
- Code
- Build circuit
- Debug
- Succeed!



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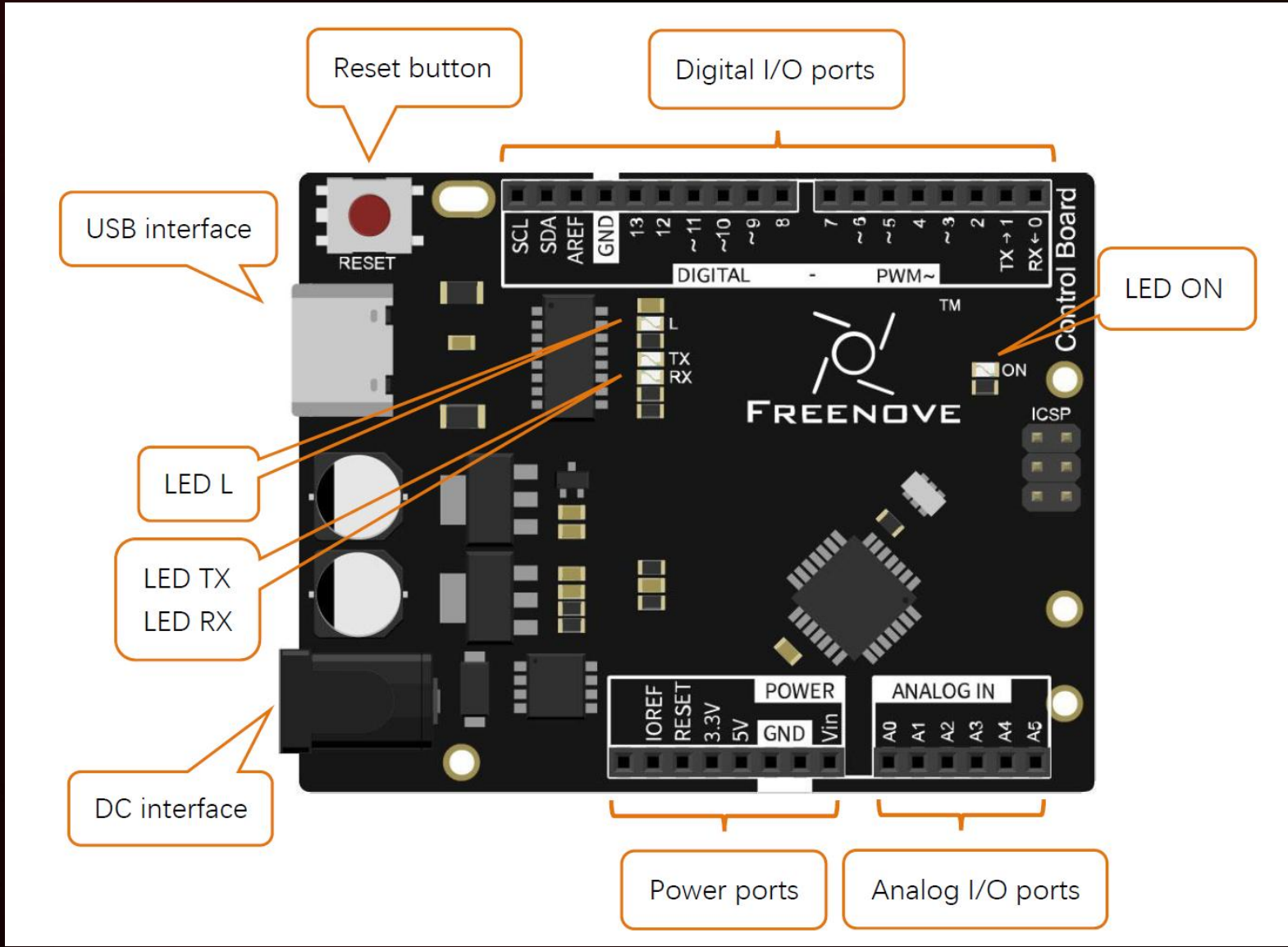
# Help & Guidebook

# The Arduino Controller Board



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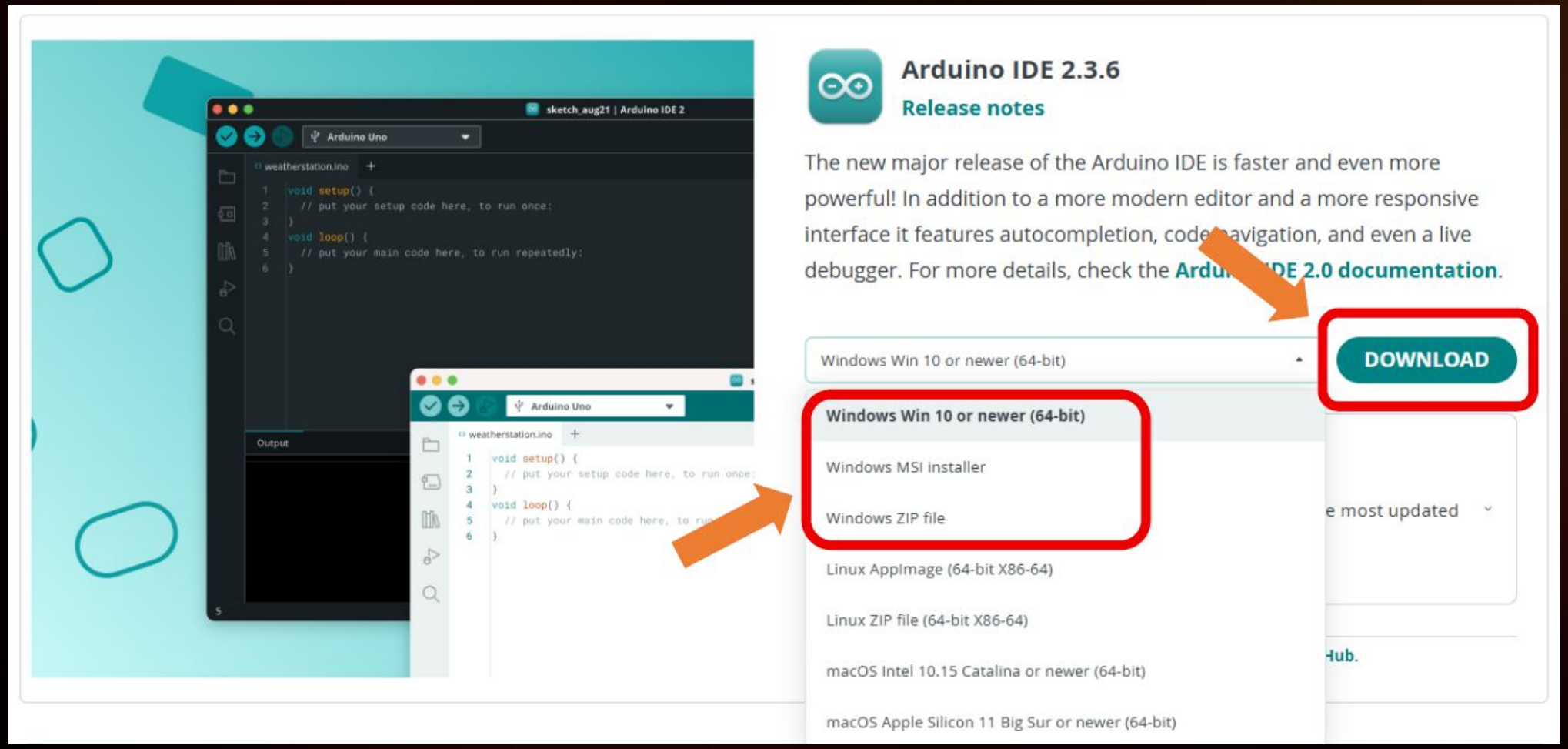
# How to program the Arduino Controller Board

- Arduino programming is based on C code
- To make adoption easy, the Arduino Kit comes hand in hand with the **Arduino Software IDE**



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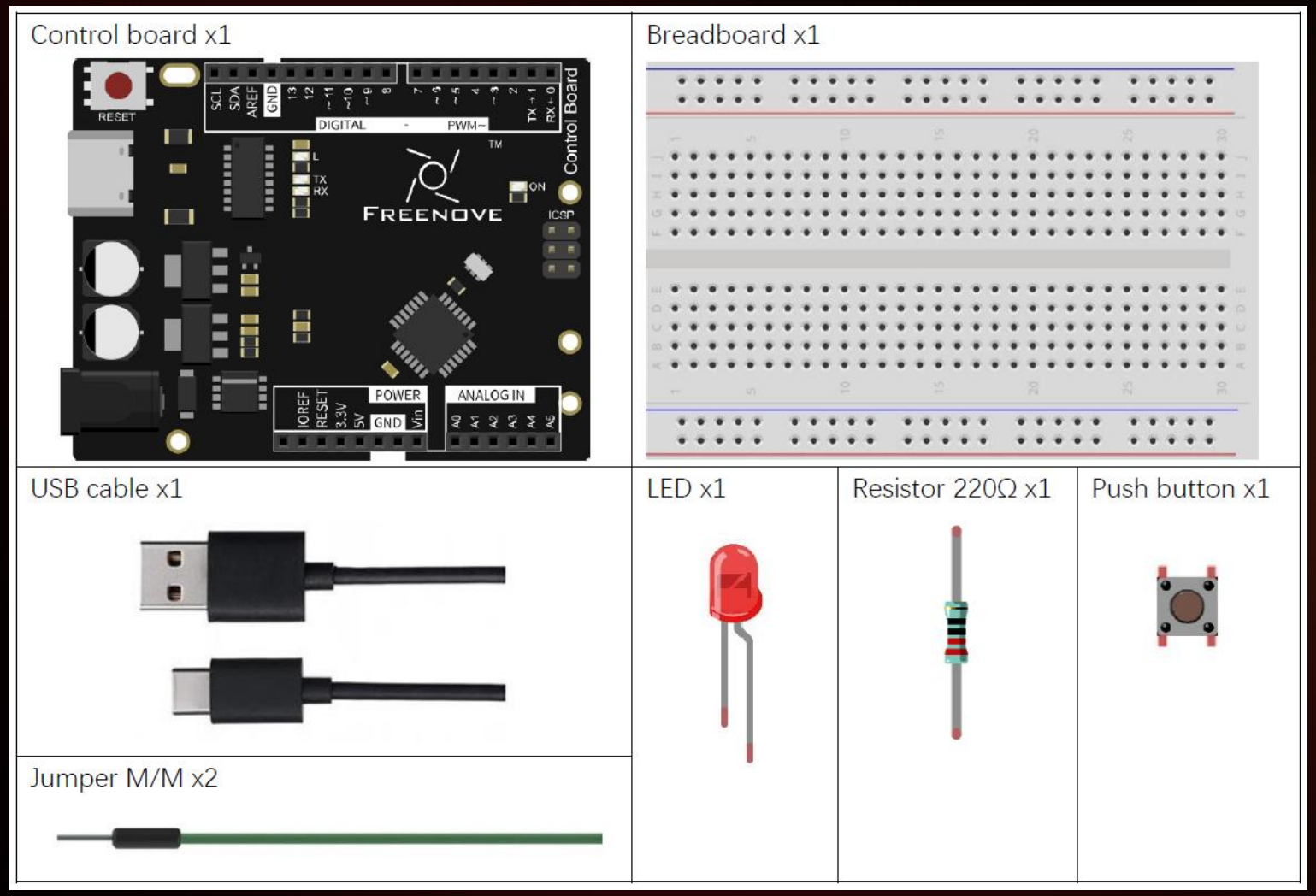
# Now, Why is that Arduino Kit box SOOOO BIG?




- The “Arduino Kit” actually comes with more than just the Controller Board!
- In the box you will find important things like the “breadboard” – every budding engineer’s BFF
- The Kit also comes with various *essential* electronic components like LEDs, Resistors, Switches, Connectors
- Also includes *advanced* components like RFID/Fob, motors, various kinds of sensors, and more!
- We use these to **BUILD OUR CIRCUIT**!!



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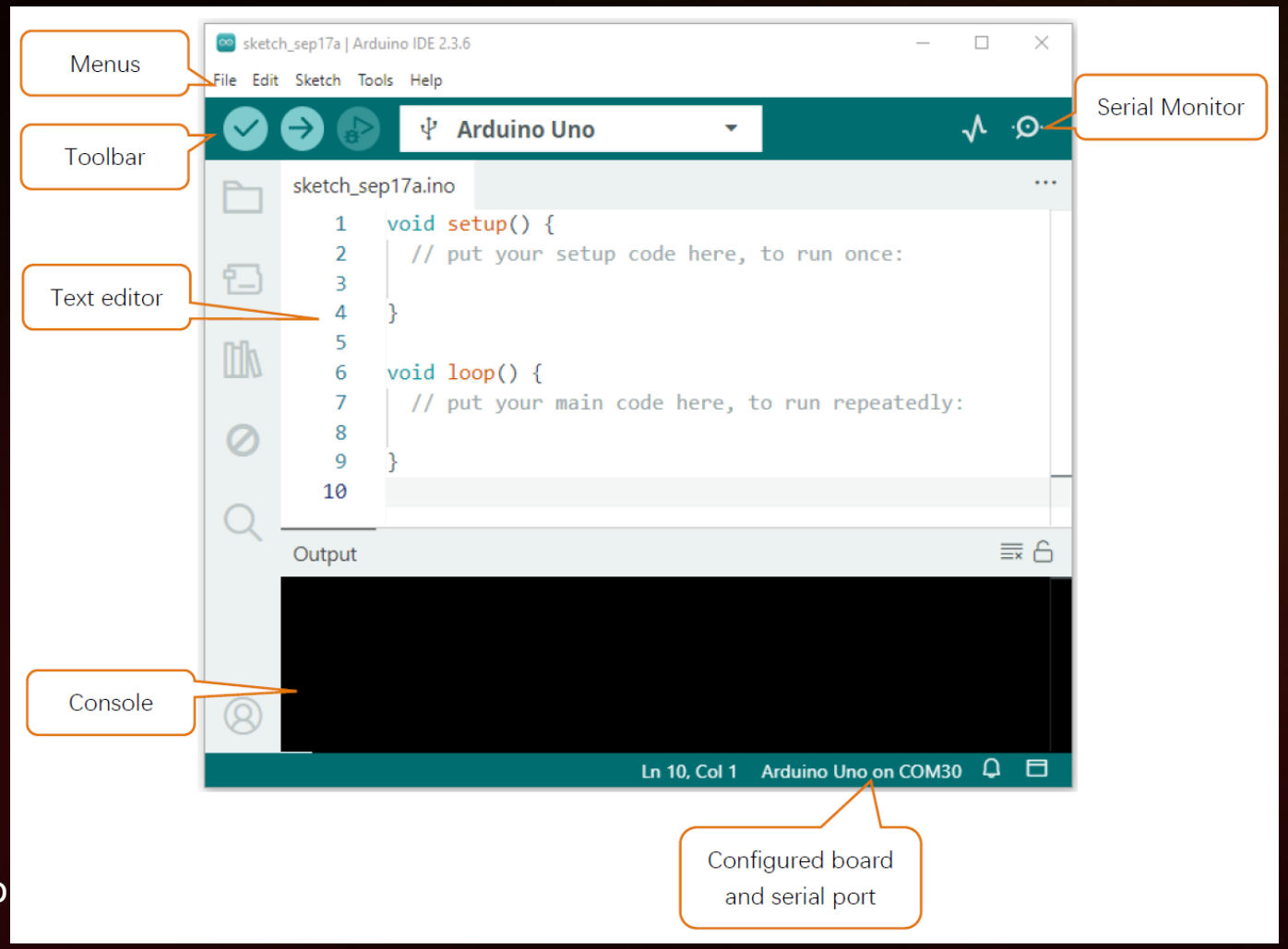
# Writing Code in the Arduino Software IDE

- Every new “sketch” in the IDE opens with a template for the “setup()” and the “loop()” functions
- The IDE editor window comes with a basic set of components along with the coding template
- The Toolbar:
  - The “check mark” button is to compile your code
  - The “right arrow” button is to “load” your compiled program onto the Arduino Controller Board
- The Console:
  - Shows the compiler output
  - Especially useful to “debug” errors in your code
- Notice:
  - We must connect the Arduino Controller Board to our computer using the USB/USB-C cord to “load” our program into the board



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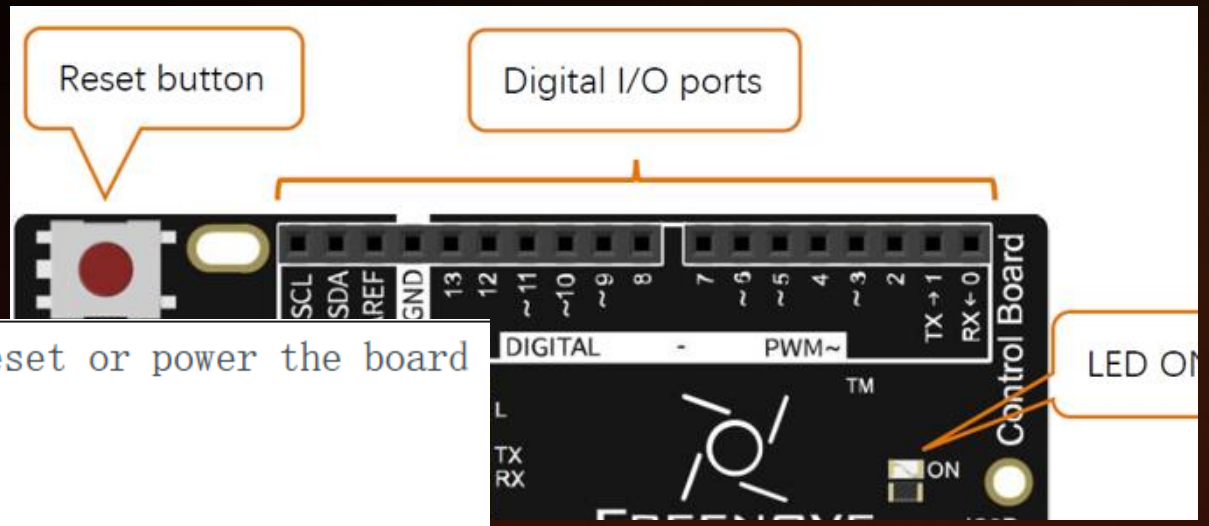
# What exactly goes into the setup() function???

- Pin programming schemes
  - The controller board provides MANY pins
  - 0-13 to be exact!
  - Plus some other pins too!




```
// the setup function runs once when you press reset or power the board
void setup() {
  // initialize digital pin 13 as an output
  pinMode(13, OUTPUT);
}
```

- In the setup() function, we “configure”
  - Which pins we want to use for this program, AND whether to configure them as:
    - INPUT: that the controller board would use to “collect” data from our **circuit**
    - OUTPUT: that the controller board would “drive” data onto



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# And just what about the loop() function???

- Pin programming schemes
  - In the loop() function
  - We tell the controller how exactly we want the pins to behave!

```
6
7   // the loop function runs over and over again forever
8   void loop() {
9     digitalWrite(13, HIGH);   // turn the LED on (HIGH is the voltage level)
10    delay(1000);              // wait for a second
11    digitalWrite(13, LOW);    // turn the LED off by making the voltage LOW
12    delay(1000);              // wait for a second
13  }
```



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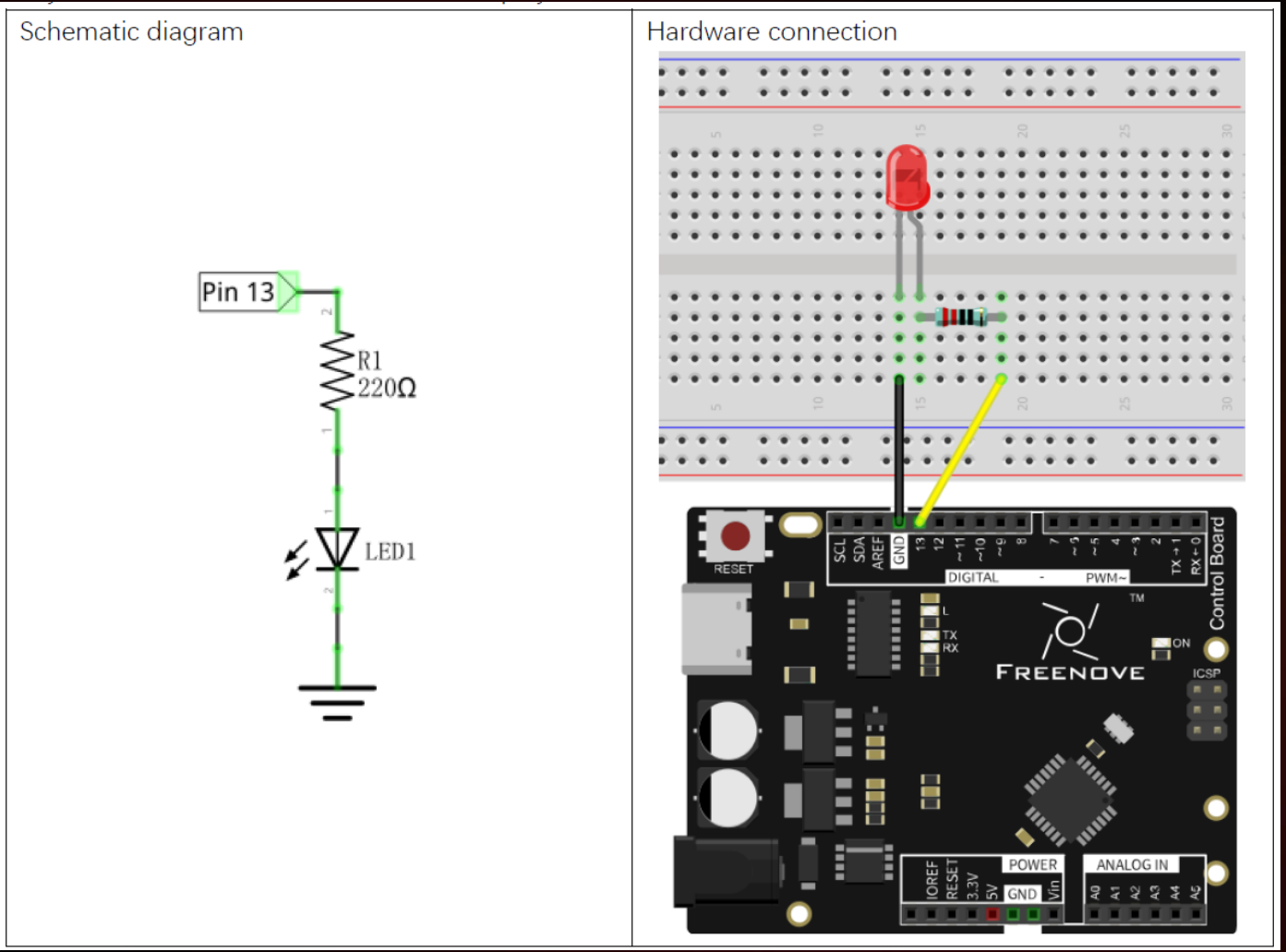
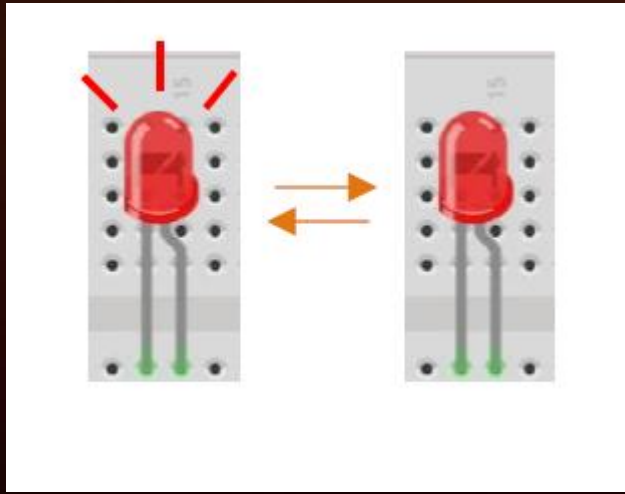
# Wait, but what do these pins ultimately even do???




- The “**Circuit**”
  - We build a circuit – one that we specifically want our program to control
  - We connect the pins to our circuit
  - And we let our program “run”





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# Whoa, how do I use that so-called breadboard???

- I’m gonna need some serious “circuit-building” training!!!

- Don’t Worry! We got you…

- Ask us for the “Circuit Knowledge” tutorial snippet!

ALSO on the web:

https://docs.freenove.com/projects/fnk0034/en/latest/



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# THANK YOU!

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